

# MSDM5004 Spring 2026

## Homework 1 Part I

Due Mar. 1

### Remarks:

(1) For all problems, write down the formulas and then calculate the results by calculators and **do not** compute it by MATLAB or other software, unless it is specified in the problem that you are required to write a code.

(2) **Do not** directly use the MATLAB built-in functions or similar available ones, unless it is specified in the problem.

(3) When you are required to write a code, you can use MATLAB or any other programming language.

1. Consider the problem of solving the equation  $f(x) = 0$ , where

$$f(x) = 1 - 4(e^x - 1) \cos x + 2x^2 + \cos 2x.$$

(1) Write down the iteration algorithm of Newton's method, then perform 4 iterations with the starting point  $x_0 = 0.6$ .

(2) Write codes using MATLAB to solve this equation with an accuracy of  $10^{-5}$  using (i) Newton's method and (ii) the secant method.

2. (1) Find the Lagrange interpolating polynomial for these data:

|        |    |   |   |
|--------|----|---|---|
| $x$    | -1 | 0 | 3 |
| $f(x)$ | 3  | 8 | 9 |

(2) Find approximation of  $f(2)$  using the interpolating polynomial.